

SBC 201 Chapter 30 Elevators and Conveying Systems — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 4–11 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 30, source file Reference\SBC 201 2024\source_reference\Chapter_30 – ELEVATORS AND CONVEYING SYSTEMS.txt.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **51** independently checkable numeric records (**50** Verified, **1** Verify source). Flattened Table 3001.3 and the damaged 3007.8 standby token are listed in the register and are not design-release values.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, lift-engineering report, fire-engineering report, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from ASME A17.1/CSA B44, ASME A17.7/CSA B44.7, ASME A18.1, ASME A90.1, ASME B20.1, ALI ALCTV, ANSI MH29.1, ASCE 24, Chapters 7, 9, 10, 11, 16 or 24, Sections 403, 405, 406, 414 or 712–717, SBC 801, ICC A117.1, NFPA 72 §21.4, UL 1784 leakage rates, commentary examples, Figure 3007.6, or the existing chapter summary. Where Chapter 30 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications. Chapter 30 does not restate the Chapter 2 high-rise definition.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Automatic sprinkler protection is not selected. **3006.2 Item 1** recognizes **Section 903.3.1.1 or 903.3.1.2**. Fire-service-access and occupant-evacuation packages (**3007.2, 3008.2**) charge **Section 903.3.1.1 only**.
4. Occupied-floor height versus the lowest level of fire department vehicle access is unconfirmed. Section **3007** applies only **where required by Section 403.6.1**. This matrix does **not** import 403.6.1 height, car count or car capacity.
5. Storey count versus grade plane, hoistway height versus **23 m**, mixed-use podium, open parking garage, multiple levels of exit discharge, and whether occupant-evacuation elevators are elected are unconfirmed.
6. Table **3001.3** is flattened concatenated HTML with duplicated Type/Standard rows. The **3007.8** standby token reads Type 60/Class 72. Both are **Verify source**.
7. Group I-1 Condition 2, I-2 and I-3 hoistway-protection triggers, below-grade transportation-station escalator width, automotive lifts, belt manlifts, conveyors, personnel/material hoists and Group H quantity limits are omitted from project-use tables.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, height band, sprinkler branch, garage, lobby type or exception exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 30 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 30 source text.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 30 application	Required project action
Sprinkler basis	Unconfirmed: neither NFPA 13 nor NFPA 13R is assumed	3006.2 Item 1 uses 903.3.1.1 or 903.3.1.2 . 3007.2 / 3008.2 require 903.3.1.1 and prohibit sprinklers in FSAE/OEE machine rooms and hoistways. 3006.3 Item 2 smoke-partition lobby needs 13 or 13R. 3006.2 Exception 2 at LED needs 903.3.1.1	Fire engineer to lock the Chapter 9 system; do not give 13R the 3007/3008 path
Fire-service-access elevators	Occupied floor stated above 23 m; height vs lowest fire-department vehicle access unconfirmed	3007 is charged only where 403.6.1 requires FSAE. Height, two-car count and 1600 kg capacity are not in this chapter	Dimension occupied floor vs fire-access grade on the code data sheet; apply Section 10 of this matrix only after 403.6.1 is confirmed
Hoistway height	Unconfirmed per bank	3006.2 Item 5 protects high-rise hoistways more than 23 m from lowest to highest floor served, even if the building is sprinklered	Measure each bank separately; a short shuttle may fall outside Item 5
Storeys vs grade plane	Unconfirmed	3002.4 stretcher car applies where elevators serve four or more stories above or below grade plane	Count stories above and below grade plane on the section; size at least one car per floor served
Occupant-evacuation elevators	Unconfirmed design option	3008 is elective. 3008.1.2 may substitute for the 403.5.2 extra stair. Lobbies, EVACS, 13-only sprinklers and standby load all change	Record whether OEE is elected in the fire strategy before locking cores
EVACS	Unconfirmed	3008.9 requires EVACS to 907.5.2.2 if OEE is used; high-rise EVACS is otherwise outbound to Chapter 9 / 403	Fire-alarm designer to confirm EVACS if 3008 is elected
Open parking / multiple LEDs	Unconfirmed	3002.7, 3006.2 Exceptions 1–2, 3007.1 Exception 1, 3007.6.2 / 3008.6.2 LED lobby exceptions, and 3008.6.1 Exception 2 all branch on garage or discharge level	Mark garage type and every level of exit discharge on life-safety plans
Mixed use / amenity escalators	Unconfirmed	3004 applies only if escalators or moving walks are provided. Group I 3006.2 items are omitted	Freeze podium uses; do not apply I-occupancy hoistway triggers to R-2 floors

Decision / gap	Current project basis	Why it controls Chapter 30 application	Required project action
Table 3001.3 / 3007.8 token	Flattened table; FSAE standby token damaged	Controls named ASME/ALI/ANSI standards and FSAE standby classification	Verify published Table 3001.3 and 3007.8 before issuing the lift specification
NOC and fire strategy	SCD NOC and stamped fire-strategy status unconfirmed	FSAE, OEE, lobby pressurization, shunt trip and standardized keys cannot be concluded from Chapter 30 alone	Engage the qualified local/fire/lift consultants before design freeze

4. Scope, standards and accessibility

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3001.1	Chapter elevator scope	This chapter governs design, construction, installation, alteration and repair of elevators and conveying systems and their components	Any elevator or conveying system	None stated	Direct	Put the lift package on the drawing index and read 3001–3008 with the referenced standards, not as a substitute for them	Verified
3001.2	Emergency two-way communication	Provide an emergency two-way communication system with visible-text and audible modes: live interactive back-and-forth in each mode; operational whenever the elevator is operational; occupant may select text or audible mode	Elevators provided	None stated	Direct	Specify live two-way text and voice in every car; show the occupant mode-select on the lift electrical schematic	Verified
3001.3	Referenced conveying standards	Design, construction, installation, alteration, repair and maintenance shall conform to the applicable standard in Table 3001.3 and to ASCE 24 in flood hazard areas, except as otherwise provided in this code	Elevators and conveying systems	Code provisions in this chapter override where they state a different requirement	External verification	Name Table 3001.3 on the lift specification; add ASCE 24 only if the site is a flood hazard area; do not import ASME or ASCE values here	Verified
Table 3001.3	Conveying-system standard list	Flattened extract concatenates Type/Standard rows and duplicates several pairings. No type-to-standard pairing is adopted as a design-release list	Equipment type assigned in the table	Readable names include automotive lifts, belt manlifts, conveyors, elevators/escalators/dumbwaiters/moving walks/material lifts, platform/stairway/wheelchair lifts and industrial scissors lifts, but attachment is unresolved	External verification	Verify the published table before listing ASME, ALI or ANSI standards on issued specifications	Verify source
3001.4	Accessible passenger elevators	Passenger elevators required to be accessible, or to serve as part of an accessible means of egress, shall comply with Sections 1009 and 1110.8	Accessible route or accessible means of egress uses a passenger elevator	ICC A117.1 car geometry is not in this chapter	External verification	Flag every accessible and accessible-egress lift; take cab, door and control dimensions from 1009 / 1110.8 / A117.1, not from this matrix	Verified
3001.5	Elevator change in use	Change from freight to passenger, passenger to freight, or from one freight class to another shall comply with Section 8.7 of ASME A17.1/CSA B44	Existing elevator changes use or freight class	ASME 8.7 alteration rules are not imported	Conditional	If a service car is converted, send the alteration to ASME 8.7; do not restripe the car class on drawings alone	Verified

5. Hoistway enclosures and cars

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3002.1	Hoistway shaft enclosure	Elevator, dumbwaiter and other hoistway enclosures shall be shaft enclosures complying with Sections 712 and 713	Hoistway provided	712/713 ratings and opening rules are not imported	External verification	Draw hoistways as 712/713 shafts; do not copy shaft hours from this matrix	Verified
3002.1.1	Hoistway opening protectives	Openings in hoistway enclosures shall be protected as required in Chapter 7	Opening in a hoistway enclosure	Car doors and associated hoistway doors at the 3003.2 recall floor may remain open during Phase I Emergency Recall Operation	External verification	Coordinate hoistway door fire-protection ratings with Chapter 7; note the recall-floor doors-open permission on the fire-alarm cause-and-effect	Verified
3002.1.2	Opening-protective hardware	Hardware on opening protectives shall be of an approved type installed as tested, except that approved interlocks, mechanical locks and electric contacts, door and gate electric contacts and door-operating mechanisms are exempt from the fire-test requirements	Hardware on hoistway opening protectives	Listed interlock/contact/operator devices are fire-test exempt; other hardware remains in the tested assembly	Direct	Keep non-exempt hardware on the labelled door assembly; list ASME interlocks as the fire-test exemption on the door schedule	Verified
3002.2	Cars per hoistway	Where four or more elevator cars serve all or the same portion of a building, locate them in not fewer than two separate hoistways. Not more than four cars in any single hoistway enclosure	Four or more cars serve the same portion	None stated	Direct	Split banks so no shaft holds more than four cars; provide a second hoistway as soon as the fourth shared-zone car is added	Verified
3002.3	Fire emergency elevator signs	Post an approved standardized pictorial sign adjacent to each elevator call station on all floors. The sign shall read: IN CASE OF FIRE, ELEVATORS ARE OUT OF SERVICE. USE EXIT STAIRS	Elevator call station	Exception 1: not required for elevators that are part of an accessible means of egress under 1009.4 . Exception 2: not required for occupant self-evacuation elevators under 3008	Direct	Place the code-text pictograph at every hall station unless the car is a documented 1009.4 or 3008 elevator	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3002.4	Ambulance stretcher elevator	Where elevators are provided in buildings four or more stories above, or four or more stories below , grade plane, provide not fewer than one elevator for fire-department emergency access to all floors. Car shall accommodate a 600 mm by 2100 mm stretcher with not less than 125 mm radius corners, horizontal open position. Identify with the star of life not less than 75 mm high, inside on both sides of the hoistway door frame	Elevators already provided; building reaches the four-story trigger	This section does not require elevators to be installed. Each floor must be served by a qualifying car; it need not be the same car for the full height	Direct	Size at least one car per floor to the stretcher envelope, place 75 mm stars on both jambs, and keep commentary 1600/1800 kg examples out of the car schedule	Verified
3002.5	Blind-hoistway emergency doors	A single blind hoistway, or an elevator on the outside of a building, shall have an emergency door in the blind portion or blank face in accordance with ASME A17.1/CSA B44	Single-car blind hoistway or exterior blank face	ASME opening size and location are not imported. Does not apply to multi-car blind hoistways	Conditional	Identify express/blind rises; provide ASME emergency doors on single-car blinds and exterior blank faces	Verified
3002.6	Additional hoistway-access doors	Doors other than hoistway doors and the elevator car door are prohibited at the point of access to an elevator car unless readily openable from the car side without a key, tool, special knowledge or effort	Extra door in front of the hoistway opening	Typical use is 3006.3 Item 3 additional doors or security doors	Direct	If a second door is used for smoke/security, specify car-side egress hardware with no key or tool	Verified
3002.7	Elevator-stair common enclosure	Elevators shall not be in a common shaft enclosure with a stairway	Elevator and stair in the same shaft	Exception: elevators within open parking garages need not be separated from stairway enclosures	Direct	Keep tower lift shafts independent of exit stairs; apply the garage exception only to a qualifying open parking garage	Verified
3002.8	Glass in elevator enclosures	Glass in elevator enclosures shall comply with Section 2409.2	Glazed hoistway or car enclosure	2409.2, 2409.3 vision panels and 2409.4 car glass are not imported	External verification	Send glazed hoistway/car details to Chapter 24; do not invent impact or marking sizes here	Verified
3002.9	Hoistway plumbing and mechanical	Plumbing and mechanical systems shall not be located in an elevator hoistway enclosure	Services routing near a hoistway	Exception: floor drains, sumps and sump pumps at the base of the hoistway, indirectly connected to the plumbing system	Direct	Keep domestic water, drainage (except the listed pit sump) and HVAC mains out of the shaft; show an indirect pit-sump connection if provided	Verified

6. Emergency operations and standby power

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3003.1	Elevator standby-power operation	Where standby power is required or furnished to operate an elevator, operation shall comply with 3003.1.1 through 3003.1.4	Standby power is required by another section or is furnished	This section does not itself require standby power. High-rise 403.4.8.2 , accessible-egress 1009.4 , FSAE 3007.8 and OEE 3008.8 are outbound chargers	External verification	Once the charger is confirmed, apply 3003.1.1–3003.1.4; do not size the generator from this clause	Verified
3003.1.1	Manual standby-power transfer	Standby power shall be manually transferable to all elevators in each bank	Standby power connected to elevators	Manual transfer does not require all cars to run simultaneously	Direct	Provide a firefighters' selector so every car in the bank can be placed on standby	Verified
3003.1.2	Single-elevator standby transfer	Where only one elevator is installed, it shall automatically transfer to standby power within 60 seconds after failure of normal power	Building with a single elevator on standby	None stated	Conditional	If a one-car scheme is used, set automatic transfer at 60 seconds	Verified
3003.1.3	Multi-elevator standby sequencing	Where two or more elevators share a common operating system and standby capacity can run all cars, all shall transfer automatically within 60 seconds . If capacity cannot run all cars, transfer in sequence, return all to the designated landing, disconnect, then keep not less than one elevator operable on standby	Two or more elevators on a common operating system with standby power	3007.8 / 3008.8 may be more restrictive for FSAE/OEE and are not relaxed by this sequencing permission	Direct	Show either full-bank 60-second transfer or sequential recall plus one remaining car; do not use sequencing to cut FSAE/OEE standby	Verified
3003.1.4	Machine-room standby venting	Where standby power is connected to elevators, machine-room ventilation or air conditioning shall be connected to the standby power source	Standby power serving elevators	None stated	Direct	Put machine-room HVAC on the same standby bus as the cars it serves	Verified
3003.2	Firefighter emergency operation	Elevators shall be provided with Phase I emergency recall and Phase II emergency in-car operation in accordance with ASME A17.1/CSA B44	Elevators provided	ASME exceptions (unrated hoistway, limited rise, LU/LA) are not imported	External verification	Specify Phase I recall and Phase II in-car operation on every car; take ASME exceptions from A17.1, not from commentary	Verified
3003.3	Standardized fire-service keys	All elevators shall be equipped to operate with a standardized fire-service elevator key in accordance with SBC 801	Elevators provided	SBC 801 key type is not imported	External verification	Call for the SBC 801 standardized fire-service key on the lift specification; do not invent a keyway here	Verified

7. Escalators and moving walks

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3004.1	Conveying-system charging	Escalators, moving walks, conveyors, personnel hoists and material hoists shall comply with 3004.2 through 3004.4	Those conveying systems are provided	Conveyors, personnel hoists and material hoists are omitted from this deliverable as not typical	Conditional	Apply this section only if amenity or podium escalators/moving walks are confirmed	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3004.2	Escalator and moving-walk materials	Construct escalators and moving walks of approved noncombustible and fire-retardant materials	Escalator or moving walk provided	Does not apply to electrical equipment, wiring, wheels, handrails, or 0.9 mm wood veneers on balustrades backed with noncombustible materials	Conditional	Specify noncombustible/fire-retardant structure; if a wood balustrade veneer is used, cap it at 0.9 mm over noncombustible backing	Verified
3004.2.1	Escalator floor-opening enclosure	Escalator floor openings shall be enclosed with shaft enclosures complying with Section 713	Escalator penetrates a floor	712.1.3 alternatives are outbound and are not imported, including commentary draft-curtain depth	External verification	Enclose escalator wells as 713 shafts, or document a 712.1.3 method from Chapter 7 — not from this matrix	Verified

8. Machine rooms, control rooms and machinery spaces

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3005.1	Machine-room access	An approved means of access shall be provided to elevator machine rooms, control rooms, control spaces and machinery spaces	Those rooms/spaces exist	Roof-access stair rules in 1009.14 are not imported	Direct	Show a building-official-approvable access route to every machine/control space on the core plans	Verified
3005.2	Machine-room independent venting	Machine rooms, machinery spaces that contain the driving machine, and control rooms or spaces that contain the operation or motion controller shall have an independent ventilation or air-conditioning system capable of maintaining temperatures within the range established for the elevator equipment	Temperature-sensitive elevator equipment in those rooms/spaces	Equipment temperature/humidity setpoints are tagged per ASME, not this chapter	Direct	Provide dedicated HVAC to machine/control spaces and set it to the lift manufacturer's tagged range	Verified
3005.3	Machine-room pressurization	A machine room, control room or control space with openings into a pressurized elevator hoistway shall be pressurized upon activation of a heat or smoke detector in that room or space	Hoistway is pressurized and the machine/control room opens into it	Pressurization of the room is required only when those detectors activate	Conditional	If 909.21 hoistway pressurization is used, extend pressurization to connected machine/control rooms and trip it from detectors in those rooms	Verified
3005.4	Machine-room fire enclosure	Machine rooms, control rooms, control spaces, and machinery spaces outside the hoistway shall be enclosed with Section 707 fire barriers or Section 711 horizontal assemblies, or both. Rating not less than the required rating of the hoistway served. Openings protected at not less than the hoistway-door fire-protection rating	Those rooms/spaces provided	707/711/hoistway hours are outbound	External verification	Match machine-room FRR to the served hoistway; take the hour rating from 713, not from this clause	Verified
3005.4 Ex. 1	Non-abutting 1-hour reduction	For other than FSAE and occupant-evacuation elevators, where those rooms/spaces do not abut and have no openings to the hoistway they serve, the enclosure may be reduced to a 1-hour fire-resistance rating	Non-abutting, no-opening machine/control space; not FSAE/OEE	Do not use for FSAE or OEE machine rooms	Conditional	Apply the 1-hour cut only on documented remote rooms that do not serve FSAE/OEE	Verified
3005.4 Ex. 2	Low-rise unrated machine rooms	For other than FSAE and OEE, in buildings four stories or less above grade plane, non-abutting rooms/spaces with no openings to the hoistway need not be fire-resistance rated	Building ≤ four stories above grade plane; room does not abut or open to the hoistway; not FSAE/OEE	Not a tower-shaft permission	Conditional	Use only for a qualifying low podium/annex; keep tower machine rooms rated	Verified
3005.5	Elevator shunt trip	Where hoistways, machine rooms, control rooms and control spaces containing elevator control equipment are protected with automatic sprinklers, provide a means per NFPA 72 Section 21.4 to disconnect main-line power automatically before water applies. The means shall not be self-resetting. Sprinklers outside those spaces shall not disconnect main-line power	Sprinklers installed in those elevator spaces	3007.4 and 3008.4 prohibit shunt trip on FSAE and OEE	Conditional	If sprinklers are in a non-FSAE/OEE machine room or hoistway, show a non-self-resetting NFPA 72 §21.4 power disconnect; do not trip from floor sprinklers outside those rooms	Verified
3005.6	Equipment-room plumbing prohibition	Plumbing systems shall not be located in elevator equipment rooms	Elevator equipment rooms	None stated	Direct	Keep water and drainage piping out of machine/control rooms	Verified

9. Hoistway opening protection

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3006.1	Hoistway and lobby charging	Provide hoistway-opening protection and enclosed elevator lobbies as listed: 3006.2 protection via 3006.3 ; underground-building lobbies via 405.4.3 ; area-of-refuge lobby via 1009.6 ; FSAE lobbies via 3007.6 ; OEE lobbies via 3008.6	Elevator hoistway openings / enclosed lobbies	405.4.3, 1009.6, 3007.6 and 3008.6 values are in those sections, not here	Direct	Pick the governing lobby package per bank (3006 vs 3007 vs 3008 vs underground/AOR) and do not mix the hour ratings	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3006.2	Hoistway-protection trigger	Protect hoistway door openings per 3006.3 where the hoistway connects more than three stories , is required to be a shaft per 712.1.1 , and any listed condition applies, including: building not sprinklered throughout per 903.3.1.1 or 903.3.1.2 ; or high-rise with hoistway height more than 23 m measured from lowest to highest floor served	Hoistway connecting more than three stories in a required shaft	Group I-1 Condition 2, I-2 and I-3 items are omitted (not typical). Sprinklered high-rise hoistways more than 23 m still hit Item 5	Direct	Protect typical tower banks under Item 5; measure hoistway height per bank, not building height	Verified
3006.2	Hoistway-protection exceptions	Protection is not required where the elevator serves only open parking garages per 406.5 ; or at the level(s) of exit discharge if that level is sprinklered per 903.3.1.1 ; or on levels where the hoistway opens to the exterior	Qualifying garage, LED, or exterior-opening level	Exception 2 requires 903.3.1.1 at the LED, not 13R. Multiple LEDs may each use the exception	Conditional	Drop 3006.3 protection only at documented garage-only, 13-sprinklered LED, or exterior-opening doors	Verified
3006.2.1	Rated-corridor hoistway protection	Where corridors are required to be fire-resistance rated per 1020.2 , protect elevator hoistway openings per 3006.3	Rated corridor required by 1020.2	1020.2 hours and occupant-load triggers are not imported	External verification	If Chapter 10 requires a rated corridor, protect hoistway openings with a 3006.3 method; do not copy corridor hours here	Verified
3006.3	Hoistway-protection methods	Where 3006.2 requires protection, provide one of: (1) enclosed lobby with Section 708 fire partitions, corridor-type 716.2.2.1 doors, and 717.5.4.1 duct/air-transfer protection; (2) enclosed lobby with Section 710 smoke partitions where the building is sprinklered throughout per 903.3.1.1 or 903.3.1.2 , plus listed smoke-partition door and 717.5.4.1 penetration rules; (3) additional doors per 3002.6 meeting 716.2.2.1.1 smoke and draft control, tested to UL 1784 without an artificial bottom seal; (4) hoistway pressurization per 909.21	3006.2 protection is required	Option 2 needs 13 or 13R throughout. 708/710/716/717/909.21 numeric criteria are not imported. FSAE/OEE lobbies are more restrictive and are not satisfied by picking a looser 3006.3 option	Direct	Select and draw one 3006.3 method per protected opening; do not use Option 2 until the sprinkler standard is locked; do not substitute this menu for 3007.6 / 3008.6	Verified
3006.4	Elevator-lobby means of egress	Elevator lobbies shall have at least one means of egress complying with Chapter 10. Egress through an enclosed elevator lobby is permitted in accordance with Item 1 of Section 1016.2	Enclosed elevator lobby	Chapter 10 widths and 1016.2 continuity rules are not imported	External verification	Give every enclosed lobby a Chapter 10 way out; if occupants pass through the lobby, check 1016.2 Item 1 so the lobby is not the only path from a rated corridor to a stair	Verified

10. Fire service access elevators

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3007.1	FSAE applicability	Where required by Section 403.6.1 , every floor above and including the lowest level of fire-department vehicle access shall be served by fire-service-access elevators complying with 3007.1 through 3007.9 , and otherwise with this chapter and ASME A17.1/CSA B44	403.6.1 requires FSAE	Exception 1: elevators that only serve an open or enclosed parking garage and the building lobby. Exception 2: not required to serve a top floor used only for building-systems equipment. 403.6.1 height, car count and 1600 kg are not imported	Conditional	Apply this section only after 403.6.1 is confirmed; serve every fire-access floor except the listed garage-only and equipment-only exceptions	Verified
3007.2	FSAE sprinkler standard	The building shall be equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 , except as otherwise permitted by 903.3.1.1.1 and as prohibited by 3007.2.1	FSAE provided	903.3.1.2 / NFPA 13R is not listed. 903.3.1.1.1 exceptions are outbound	Conditional	Specify NFPA 13 / 903.3.1.1 throughout if FSAE is used; do not substitute 13R	Verified
3007.2.1	FSAE sprinkler-free rooms	Automatic sprinklers shall not be installed in machine rooms, elevator machinery spaces, control rooms, control spaces and elevator hoistways of fire-service-access elevators	FSAE machine/control rooms and hoistways	None stated	Conditional	Omit sprinklers from FSAE machine rooms, control spaces and hoistways on the sprinkler drawings	Verified
3007.2.2	FSAE floor-flow monitoring	Sprinkler control-valve supervisory switch and water-flow-initiating device for each floor , monitored by the building fire-alarm system	FSAE building sprinkler system	None stated	Conditional	Show per-floor valve tamper and flow on the fire-alarm matrix for every FSAE-served floor	Verified
3007.3	FSAE hoistway water protection	Water from sprinklers outside the enclosed lobby shall be prevented from infiltrating the hoistway enclosure by an approved method	FSAE hoistway opening	Protects against sprinkler water, not fire-hose streams. Method is performance-based (AHJ-approved)	Conditional	Detail an approved threshold/drain/curb/gasket strategy at the FSAE lobby or hoistway opening	Verified
3007.4	FSAE shunt-trip prohibition	Means for elevator shutdown in accordance with 3005.5 shall not be installed on fire-service-access elevator systems	FSAE	None stated	Conditional	Delete shunt trip from FSAE controllers even if other cars keep 3005.5	Verified
3007.5–3007.5.1	FSAE hoistway enclosure	FSAE hoistway shall be a Section 713 shaft enclosure. Structural integrity shall comply with Sections 403.2.2.1 through 403.2.2.4	FSAE hoistway	713 hours and 403.2.2 impact levels are not imported	External verification	Draw the FSAE shaft as a 713 enclosure and send impact integrity to 403.2.2; do not copy those values here	Verified
3007.5.2	FSAE hoistway lighting	When firefighters’ emergency operation is active, illuminate the entire hoistway height at not less than 11 lux , measured from the top of the car of each FSAE	FSAE on firefighters’ emergency operation	None stated	Conditional	Specify hoistway lighting that delivers 11 lux at the car top for the full rise when Phase I/II is active	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3007.6	FSAE enclosed lobby	Each FSAE shall open into an enclosed fire-service-access elevator lobby complying with 3007.6.1 through 3007.6.5 . Egress through the lobby is permitted per 1016.2 Item 1	FSAE landing	Exception: where the FSAE has two entrances on a floor, the second entrance may be protected per 3006.3	Conditional	Provide a 3007.6 lobby at the primary FSAE entrance; a second entrance may use a 3006.3 method	Verified
3007.6.1	FSAE lobby stair access	The enclosed FSAE lobby shall have direct access to an interior exit stairway or ramp enclosure	FSAE lobby	Exception: access may be through a protected path with fire protection not less than the lobby enclosure, separated from the lobby by a 716.2.2.1 smoke and draft control assembly	Conditional	Place an exit stair on the FSAE lobby or document an equally protected path with a smoke/draft door	Verified
3007.6.2	FSAE lobby enclosure	Enclose the FSAE lobby with a smoke barrier having a fire-resistance rating of not less than 1 hour , except lobby doorways per 3007.6.3	FSAE lobby	Exception: enclosed FSAE lobbies are not required at the levels of exit discharge	Conditional	Draw a 1-hour smoke-barrier lobby on every FSAE floor except LED	Verified
3007.6.3	FSAE lobby doorways	Other than doors to the hoistway, elevator control room or control space, each doorway to the enclosed FSAE lobby shall have a 3/4-hour fire door assembly complying with Section 716 , meeting 716.2.2.1.1 smoke and draft control, tested to UL 1784 without an artificial bottom seal	Doorway into the FSAE lobby enclosure	Hoistway, control-room and control-space doors are excluded from this doorway rule. 716 leakage limits are not imported	Conditional	Specify 3/4-hour smoke/draft lobby doors tested without a bottom seal; keep 20-minute smoke-barrier defaults off this door type	Verified
3007.6.4	FSAE lobby size	Regardless of the number of FSAE cars served by the same lobby, the enclosed lobby shall be not less than 14 m² with a dimension of not less than 2.4 m	FSAE lobby	None stated	Conditional	Dimension every FSAE lobby to 14 m² min and 2.4 m min width; do not enlarge it by car count	Verified
3007.6.5	FSAE identification symbol	Install the standardized FSAE pictorial symbol on each side of the hoistway door frame, on the portion at right angles to the FSAE lobby, per Figure 3007.6.5. Symbol not less than 75 mm high; helmet contrasts with the background; vertical centerline centered on the frame; each symbol not less than 1.9 m and not more than 2 m above the finished floor at the threshold	FSAE hoistway door frame at the lobby	Figure 3007.6.5 graphic is not reproduced here	Conditional	Place a 75 mm contrasting fire-hat symbol on both FSAE jamb returns at 1.9–2 m AFF	Verified
3007.7	FSAE system monitoring	Continuously monitor the FSAE at the fire command center by a standard emergency-service interface meeting NFPA 72	FSAE	NFPA 72 interface layout is not imported	External verification	Show FSAE status at the fire command center on an NFPA 72 emergency-service interface; do not design the annunciator from this clause	Verified
3007.8	FSAE standby-power features	The following serving each FSAE shall be supplied by both normal power and standby power: elevator equipment; elevator hoistway lighting; ventilation/cooling for machine rooms, control rooms, machine spaces and control spaces; elevator car lighting	Each FSAE	None stated for the feature list	Conditional	Put FSAE equipment, hoistway lighting, machine-room HVAC and car lighting on normal plus standby	Verified
3007.8	FSAE standby classification token	Published extract reads standby supply as Type 60/Class 72 . The token is not an unambiguous Type/Class/Level designation. No standby classification is adopted	FSAE standby supply	Do not substitute Type 60/Class 2/Level 1 from 3008.8 or from memory	Conditional	Hold the FSAE generator classification until the published 3007.8 token is verified	Verify source
3007.8.1	FSAE wiring protection	Wires or cables outside the hoistway and machine room that provide normal or standby power, control signals, car communication, lighting, HVAC, ventilation and fire-detecting systems to FSAE shall be protected by one of: UL 2196 cables with fire-resistance rating not less than 2 hours ; listed electrical circuit protective systems not less than 2 hours ; or construction not less than 2 hours	FSAE feeder/control/communication cables outside the hoistway and machine room	Exception: wiring and cables to control signals need not be protected if they do not serve Phase II emergency in-car operations	Conditional	Route FSAE life-safety cables in a 2-hour method; exempt only non-Phase-II control-signal cables	Verified
3007.9–3007.9.1	FSAE standpipe hose connection	A Class I standpipe hose connection per Section 905 shall be provided in the interior exit stairway or ramp that has direct access from the enclosed FSAE lobby. That exit enclosure shall have access to the floor without passing through the enclosed FSAE lobby	FSAE lobby / associated exit enclosure	905 hose-valve criteria are not imported	External verification	Place the Class I hose valve in the FSAE-adjacent stair and provide a floor-access door that does not require opening the lobby	Verified

11. Occupant evacuation elevators

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3008.1	Occupant-evacuation elevator option	Elevators used for occupant self-evacuation during fires shall comply with 3008.1 through 3008.10	Designer elects occupant-evacuation elevators	Not mandatory unless chosen. 403.6.2 charging is outbound	Conditional	Apply this section only if OEE is written into the fire strategy	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3008.1.1	Occupant-evacuation elevator count	Number of OEE cars from an egress analysis of one scenario: full-building evacuation in less than 1 hour , or evacuation of the five consecutive floors with the highest cumulative occupant load in less than 15 minutes. Not less than one elevator in each bank shall be designated. Not less than two shall be provided in each OEE lobby where more than one elevator opens into the lobby. Provide signage denoting which elevators are available	OEE elected	Analysis method is performance-based; car speeds and handling capacity are not in this chapter	Conditional	Run the 1-hour or 15-minute / five-floor analysis, keep at least one OEE per bank and two where a lobby has multiple openings, and sign the designated cars	Verified
3008.1.2	Additional-stair substitution	Where an additional means of egress is required by Section 403.5.2 , an additional exit stairway is not required in buildings provided with occupant-evacuation elevators complying with 3008.1	OEE provided and 403.5.2 would otherwise require an extra stair	403.5.2 height trigger and R-2 extra-stair exception are not imported	Conditional	If OEE is elected as the 403.5.2 alternative, record that substitution in the fire strategy; do not delete stairs required by Chapter 10	Verified
3008.1.3–3008.1.4	Occupant-evacuation operation plan	The building shall have an approved fire-safety and evacuation plan per Section 404 of SBC 801 incorporating occupant-evacuation elevator procedures. Use the elevators in accordance with occupant-evacuation operation in ASME A17.1/CSA B44 and that plan	OEE elected	SBC 801 plan content and ASME 2.27.11 operation are not imported	External verification	Include OEE procedures in the SBC 801 fire-safety plan and specify ASME occupant-evacuation operation	Verified
3008.2–3008.2.2	Occupant-evacuation sprinklers	Building equipped throughout with an approved, electrically supervised sprinkler system per 903.3.1.1 , except as permitted by 903.3.1.1.1 and as prohibited by 3008.2.1 . No sprinklers in OEE machine rooms, machinery spaces, control rooms, control spaces and hoistways. Per-floor valve supervisory switch and water-flow device monitored by the fire-alarm system	OEE elected	903.3.1.2 / NFPA 13R is not listed	Conditional	Use 903.3.1.1 throughout, omit sprinklers from OEE machine/hoistway spaces, and monitor flow/tamper per floor	Verified
3008.3	Occupant-evacuation water protection	Water from sprinklers outside the enclosed lobby shall be prevented from infiltrating the hoistway enclosure by an approved method	OEE hoistway opening	Same performance intent as 3007.3	Conditional	Detail an approved water-infiltration control at each OEE lobby/hoistway opening	Verified
3008.4	Occupant-evacuation shunt-trip prohibition	Means for elevator shutdown in accordance with 3005.5 shall not be installed on occupant-evacuation elevator systems	OEE	None stated	Conditional	Delete shunt trip from OEE controllers	Verified
3008.5–3008.5.1	Occupant-evacuation hoistway enclosure	OEE hoistways shall be Section 713 shaft enclosures. Structural integrity shall comply with 403.2.2.1 through 403.2.2.4	OEE hoistway	713 and 403.2.2 values are not imported	External verification	Draw OEE shafts as 713 enclosures and send impact integrity to 403.2.2	Verified
3008.6	Occupant-evacuation enclosed lobby	OEE shall open into an enclosed elevator lobby complying with 3008.6.1 through 3008.6.6 . Egress through the lobby is permitted per 1016.2 Item 1	OEE landing	None stated at this charging clause	Conditional	Provide a 3008.6 lobby at every OEE opening used for occupant self-evacuation	Verified
3008.6.1	Occupant-evacuation lobby stair access	Direct access from the enclosed OEE lobby to an interior exit stairway or ramp	OEE lobby	Exception 1: protected path not less than the lobby enclosure, separated by a 716.2.2.1 smoke and draft control assembly. Exception 2: elevators that only serve an open parking garage and the building lobby need not provide direct access	Conditional	Place an exit stair on the OEE lobby, or document the protected-path exception; garage-only cars may skip direct stair access	Verified
3008.6.2	Occupant-evacuation lobby enclosure	Enclose the OEE lobby with a smoke barrier having a fire-resistance rating of not less than 1 hour , except lobby doorways per 3008.6.3	OEE lobby	Exception: enclosed OEE lobbies are not required at the levels of exit discharge	Conditional	Draw a 1-hour smoke-barrier OEE lobby on every floor except LED	Verified
3008.6.3	Occupant-evacuation lobby doorways	Other than doors to the hoistway, machine rooms, machinery spaces, control rooms and control spaces within the lobby smoke barrier, each doorway shall have a 3/4-hour fire door assembly complying with Section 716 , meeting 716.2.2.1.1 , tested to UL 1784 without an artificial bottom seal	Doorway into the OEE lobby enclosure	Listed equipment-room doors inside the lobby are excluded	Conditional	Specify 3/4-hour smoke/draft OEE lobby doors tested without a bottom seal	Verified
3008.6.3.1–3008.6.3.2	Occupant-evacuation lobby door vision and closing	Install a fire-protection-rated vision panel complying with Section 716 in each lobby-doorway fire door, located to furnish clear vision of the OEE lobby. Each such door shall be automatic-closing upon any fire-alarm signal from the building EVACS	OEE lobby doorway	716 glazing sizes are not imported	Conditional	Put a 716 vision panel in every OEE lobby door and close those doors on any EVACS fire-alarm signal	Verified
3008.6.4	Occupant-evacuation lobby size	Lobby floor area shall accommodate, at 0.27 m² per person, not less than 25 percent of the occupant load of the floor area served, and one wheelchair space 0.75 m by 1.2 m for each 50 persons, or portion thereof, of that occupant load	OEE lobby	Exception: lobbies serving multiple banks shall have minimum floor area approved on an individual basis, consistent with the fire-safety and evacuation plan. Do not use Figure 3007.6 comparison cells	Conditional	Size each OEE lobby from the floor occupant load using 0.27 m² × 25 percent plus wheelchair spaces; obtain AHJ approval for multi-bank lobbies	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
3008.6.5	Occupant-evacuation elevator signage	Post an approved sign indicating elevators are suitable for occupant self-evacuation on all floors, adjacent to each elevator call station serving OEE	OEE hall station	Replaces the 3002.3 “do not use elevators” sign on those cars (3002.3 Exception 2)	Conditional	Place OEE-available signs at those hall stations and omit the 3002.3 pictograph on the same cars	Verified
3008.6.6	Occupant-evacuation lobby two-way communication	Provide two-way communication in each OEE lobby to the fire command center or an alternate location approved by the fire department, designed and installed per Sections 1009.8.1 and 1009.8.2	OEE lobby	1009.8 device layout is not imported	External verification	Put an area-of-refuge-type two-way station in every OEE lobby; take device rules from 1009.8	Verified
3008.7–3008.7.1	Occupant-evacuation system monitoring	Continuously monitor OEE at the fire command center or a fire-department-approved central control point, displaying car floor, direction, occupancy status, normal-power status, standby/emergency-power status, and activation of fire-alarm initiating devices in listed elevator spaces. Provide means there to manually initiate Phase I Emergency Recall per ASME A17.1/CSA B44	OEE	ASME recall wiring is not imported	Conditional	Show the six status points and a manual Phase I recall control at the fire command center	Verified
3008.8–3008.8.1	Occupant-evacuation standby power	The following serving each OEE shall be supplied by both normal power and Type 60/Class 2/Level 1 standby power: elevator equipment; ventilation/cooling for machine rooms, control rooms, machinery spaces and control spaces; elevator car lighting. Standby-power load based on the number of OEE cars from 3008.1.1	Each OEE	Unlike 3003.1.3, this clause does not permit shedding cars that are part of the OEE set. Hoistway lighting is not in this feature list	Conditional	Put OEE equipment, machine-room HVAC and car lighting on Type 60/Class 2/Level 1 standby sized to the 3008.1.1 car count	Verified
3008.8.2	Occupant-evacuation wiring protection	Wires or cables outside the hoistway, machine room, control room and control space that provide normal or standby power, control signals, car communication, lighting, HVAC, ventilation and fire-detecting systems to OEE shall be protected by one of: UL 2196 cables not less than 2 hours ; listed electrical circuit protective systems not less than 2 hours ; or construction not less than 2 hours	OEE feeder/control/communication cables outside those rooms	Exception: control-signal wiring need not be protected if it does not serve Phase II emergency in-car operation	Conditional	Route OEE life-safety cables in a 2-hour method; exempt only non-Phase-II control-signal cables	Verified
3008.9–3008.9.1	Occupant-evacuation EVACS	Provide an emergency voice/alarm communication system accessible to the fire department per Section 907.5.2.2 . Install not fewer than one audible and one visible notification appliance within each OEE lobby	OEE elected	907.5.2.2 speaker layout is not imported	External verification	Include EVACS in the fire-alarm scope and place at least one audible and one visible appliance in every OEE lobby	Verified

12. Project-use controls

1. Use **Verified** rows for initial scoping after the row trigger is confirmed (elevator actually provided; FSAE/OEE actually charged or elected).
2. Treat **Verify source** rows as hold points. Table **3001.3** pairings and the **3007.8** Type 60/Class 72 token are not design-release values.
3. Do not import ASME A17.1 car sizes, Chapter 7 shaft/door hours, 903 sprinkler densities, 909.21 pressurization criteria, 1009/A117.1 cab geometry, 403.6.1 height/car-count/1600 kg, UL 1784 leakage rates, or Figure 3007.6 comparison cells into issued drawings from this matrix.
4. Do not assume NFPA 13 versus 13R for **3006**. Do not give 13R the **3007** or **3008** path.
5. Do not apply Group I hoistway-protection items, subway escalator width, or hoist/conveyor rules to the R-2 tower unless the gap register is opened for that use.
6. Record FSAE, OEE, sprinkler-standard and hoistway-height decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

13. Coverage summary

Internal inventory of the attached Chapter 30 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 51
- **Verified:** 50
- **Verify source:** 1

Counts by top-level section

Top-level section	Records
3001	0
3002	10
3003	4
3004	2
3005	2

Top-level section	Records
3006	3
3007	12
3008	18
Table 3001.3	0 numeric cells (flattened; see register)

Figures (3002.3, 3002.4, 3002.6, 3004.2.1, 3007.6, 3007.6.5) and commentary examples were not inventoried. Figure 3007.6 lobby-comparison cells (14 m, 0.3 m) were not used.

Coverage cross-check against SBC 201 Chapter 30 Elevators and Conveying Systems (2024)_CS.md was topics-only: scope and standards; hoistway enclosures; emergency operation; conveying systems; machine rooms; hoistway opening protection; fire-service-access elevators; occupant-evacuation elevators. No CS.md value was copied into a matrix cell.

14. Unresolved-source register

Affected table / clause	Why unverified	Control note
Table 3001.3	Flattened concatenated HTML; Type/Standard header and body are merged; several pairings are duplicated	Do not issue a type-to-standard list from this extract. Verify the published table before naming ASME/ALI/ANSI standards
3007.8 standby token	Published text is Type 60/Class 72. Not an unambiguous Type/Class/Level designation	Do not repair from 3008.8 (Type 60/Class 2/Level 1) or from memory. Hold FSAE generator classification until the published clause is checked